VELS
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AI-Based Electrical Fire Prevention System Using IoT Sensors and Machine Learning

Abstract

Traditional fire alarm systems only react once smoke or flames are already present, often too late to prevent serious damage. This project introduces an AI-based Electrical Fire Prevention System that uses IoT sensors and machine learning to catch warning signs before a fire even starts. The system continuously monitors temperature, current, voltage, and smoke levels through sensors connected to an ESP32 microcontroller. Instead of relying on fixed thresholds, the readings are analyzed together by a trained machine learning model, which identifies abnormal patterns that typically precede ignition. Based on this analysis, the system classifies conditions into four levels: Safe, Warning, High Risk, or Critical. When a Critical risk is detected, the system responds automatically by cutting power via a relay, sounding a buzzer, sending an SMS alert, and logging the event. By predicting electrical fire probability using multi-sensor pattern recognition, the system provides a proactive, low-cost approach to fire prevention.

Keywords: Electrical Fire Prevention; Internet of Things; Machine Learning; ESP32 Microcontroller; Predictive Safety; Smart Sensors; Fire Risk Classification.

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